

Supporting Information

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SI Methods

Specimen Collection. The collection of all bird specimens was authorized by the local forestry administrations and was in compliance with the National Wildlife Conservation Law of China. All birds were handled in accordance with regulations of the Animal Experimental and Medical Ethics Committee of the Institute of Zoology, Chinese Academy of Sciences. Voucher specimens were cataloged in the ornithological collection of the National Zoological Museum, Institute of Zoology, Chinese Academy of Sciences.

Sample Preservation. We collected pectoral muscle from each bird specimen and preserved subsamples in RNAhold (TransGen) and 100% ethanol as sources of RNA and genomic DNA, respectively. For a subset of specimens, we also collected 20–100 μ L of whole blood from the brachial or ulnar vein using heparinized microcapillary tubes. Red blood cells were separated from the plasma fraction by centrifugation, and the packed red cells were subsequently stored at -80°C before protein purification.

PCR, Cloning, and Sequencing. For each of the 162 bird specimens, including those used as subjects in the experimental analyses of Hb function, we extracted DNA from pectoral muscle tissue using the DNeasy Tissue kit (Qiagen). We annotated globin genes in the genome assembly of *Parus humilis* (56) and designed paralog-specific primers using 5' and 3' UTR sequences. We then PCR-amplified the α^A -, α^D -, and β^A -globin genes using KD-Plus DNA polymerase (TransGen). To isolate RNA from muscle tissue, we used the Mag-BIND Total RNA kit (Omega), and we then amplified full-length cDNAs of the α^A -, α^D -, and β^A -globin genes using a One-Step RT-PCR SuperMix kit (TransGen). Following gel purification, we cloned PCR and RT-PCR products using the Blunt Simple Cloning kit (TransGen). For each species, we sequenced full-length coding regions or cDNAs of at least three to five individual specimens, and we sequenced four to six clones per gene to recover both alleles. This protocol enabled us to determine diploid genotypes for each of the three adult-expressed globin genes in each individual bird.

Protein Purification and in Vitro Analysis of Hb Function. We purified HbA and HbD from hemolysates of wild-caught birds by means of anion-exchange FPLC using an Äkta Pure system and a HiTrap Q HP column (GE Healthcare). This procedure removes endogenous organic phosphates, yielding stripped Hb samples. Using purified Hb solutions (0.3 mM heme), we measured O_2 equilibrium curves at 37°C in 0.1 M Hepes buffer (pH 7.4) in the absence (stripped) and presence of 0.1 M KCl and IHP (at twofold molar excess over tetrameric Hb) and in the simultaneous presence of KCl and IHP. For the analyses of purified native Hbs, we measured O_2 equilibria of 3- μ L thin-film samples in a modified diffusion chamber where absorption at 436 nm was monitored during stepwise changes in the equilibration of N_2/O_2 mixtures generated by precision Wösthoff gas-mixing pumps (57–59). We estimated values of P_{50} and n_{50} by fitting the Hill equation $Y = \text{PO}_2^n / (P_{50}^n + \text{PO}_2^n)$ to the experimental O_2 saturation data by means of nonlinear regression (Y = fractional O_2 saturation; n , cooperativity coefficient). The nonlinear fitting was based on five to eight equilibration steps. Free Cl^- concentrations were measured with a model 926S Mark II chloride analyzer (Sherwood Scientific Ltd). For the analysis of purified rHbs representing *AncParidae* and the wild-type genotypes of *Lophophanes dichrous* and *Parus humilis*, we used a similar

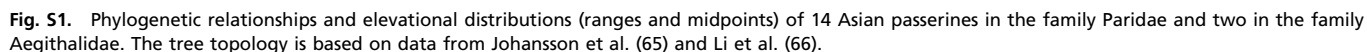
procedure using the BOBS diffusion chamber and gas-mixing system (Loligo Systems). P_{50} values were measured at three different pH levels (~ 7.0 , 7.3, and 7.5) using a pH-1 Micro pH meter and a needle-type pH microsensor (PreSens Precision Sensing GmbH). A linear regression was fit to plots of $\log P_{50}$ vs. pH, and the resulting equation was used to correct P_{50} values to pH 7.4.

Vector Construction and Site-Directed Mutagenesis. The synthesized α^A - β^A globin gene cassette was cloned into a custom pGM vector system (Promega) along with the *methionine aminopeptidase* (MAP) gene, as described previously (25, 53). We engineered each of the α - and β -chain codon substitutions via whole-plasmid amplification using mutagenic primers and Phusion High-Fidelity DNA Polymerase, phosphorylation with T4 Polynucleotide Kinase, and circularization with an NEB Quick Ligation Kit (New England BioLabs). Each engineered codon change was verified by DNA sequencing.

Expression and Purification of Recombinant Hbs. Recombinant Hbs were expressed in JM109 (DE3) *E. coli* host cells. Bacterial cells were selected in LB agar with dual antibiotics (ampicillin and kanamycin) to ensure that transformants received both pGM and pCO-MAP plasmids for expression. The expression of each rHb mutant was carried out in 1.5 L of TB medium (Research Products International). Bacterial cells were grown at 37°C in an orbital shaker at 200 rpm until absorbance values reached 0.6–0.8 at 600 nm. The bacterial cultures were induced by 0.2 mM isopropyl β -D-1-thiogalactopyranoside (IPTG) and were then supplemented with hemin (50 $\mu\text{g}/\text{mL}$) and glucose (20 g/L). The bacterial culture conditions and the protocol for preparing cell lysates are described in Natarajan et al. (53).

We purified each rHb sample by means of two-step ion-exchange chromatography as described previously (15–17, 21, 25, 26, 29, 53, 60). We then passed the samples through an anion-exchange column (HiTrap Q-XL, 17-5159-01; GE Healthcare) equilibrated with 20 mM Tris-HCl buffer (0.5 mM EDTA, pH 8.0) and eluted using a linear gradient of 0–0.5 M NaCl. We then passed the eluted fractions through a cation-exchange column (HiTrap SP-XL, 17-5161-01; GE Healthcare) equilibrated with 20 mM Hepes buffer (0.5 mM EDTA, pH 8.0) followed by elution using a linear gradient of 0–0.5 M NaCl. We desalted the samples by means of dialysis against 10 mM Hepes buffer (pH 7.4) at 4°C , and we concentrated the eluted fractions of each rHb sample by means of centrifugal filtration. We confirmed the integrity of purified rHb samples using SDS-PAGE and IEF. After preparing rHb samples as oxygenated Hb, deoxygenated Hb, and carbonmonoxy derivatives, we measured absorbance at 450–600 nm to confirm that the absorbance maxima matched those of the native HbA samples.

Phylogeny Estimation. We aligned the full globin gene sequences (exons + introns) using MUSCLE version 3.5 (61) and verified that exon length was conserved. In all cases we estimated maximum likelihood (ML) trees in MEGA version 7.0 (62) using a GTR+ Γ model of nucleotide substitution with five rate classes, and we evaluated support with 1,000 bootstrap pseudoreplicates. We estimated Bayesian trees in MrBayes version 3.2 (63) using default priors, running four simultaneous chains for 2×10^7 generations and sampling trees every 1,000 generations. We assessed convergence by measuring the SD of the split frequency among parallel chains, using a split frequency of <0.01 as the criterion for convergence. We discarded trees collected before chains converged.



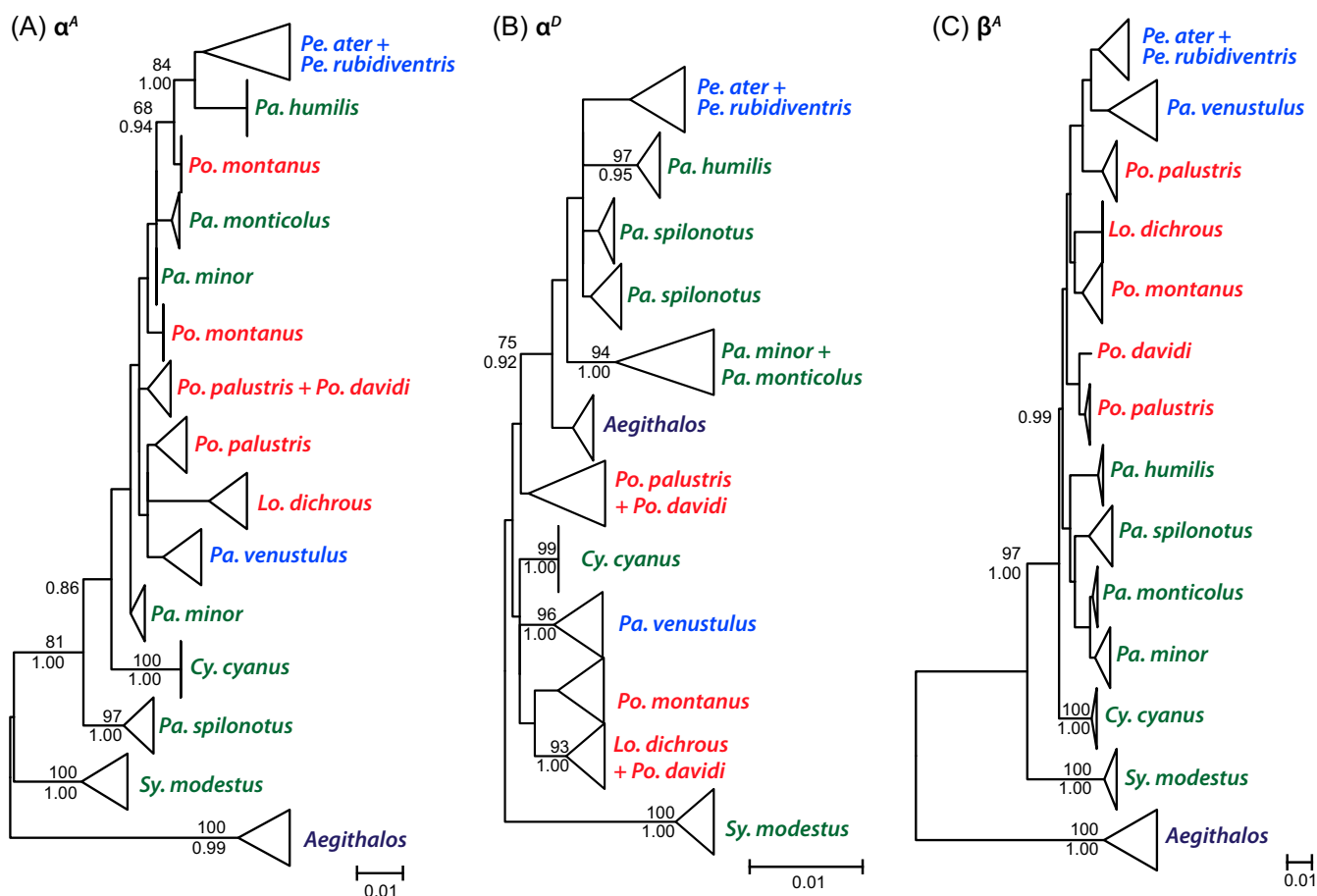
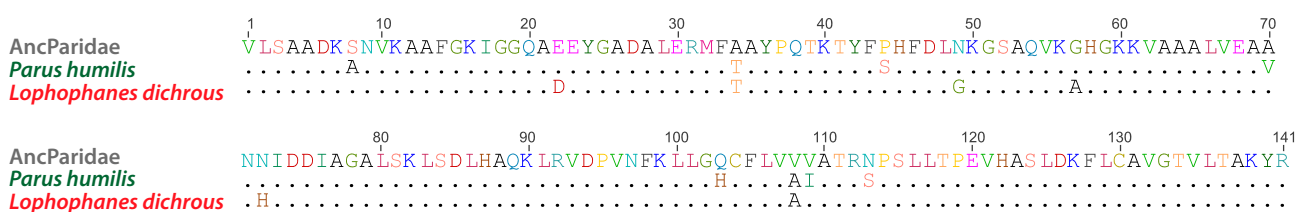


Fig. S2. Phylogenies of α^A -globin (A), α^D -globin (B), and β^A -globin (C) genes exhibit high levels of genealogical discordance. Species names are color-coded according to the four clades depicted in the species tree (Fig. S1).

A) α^A -globin



B) β^A -globin

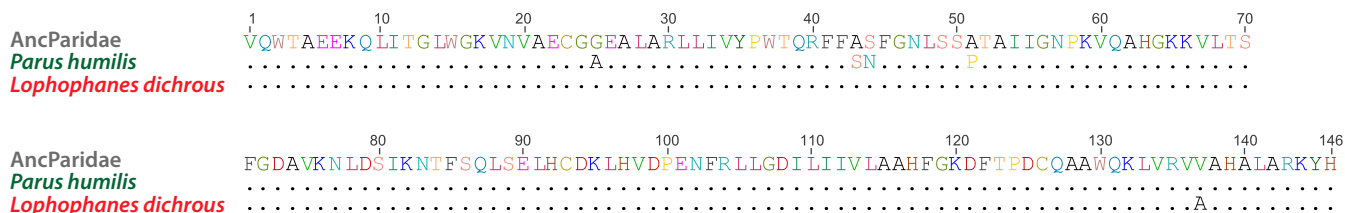


Fig. S3. Wild-type α^A -globin (A) and β^A -globin (B) sequences of *Parus humilis*, *Lophophanes dichrous*, and their reconstructed common ancestor, AncParidae. Posterior probabilities for ancestral amino acid states averaged 0.997 across 141 α -chain sites and 1.000 across 146 β -chain sites.

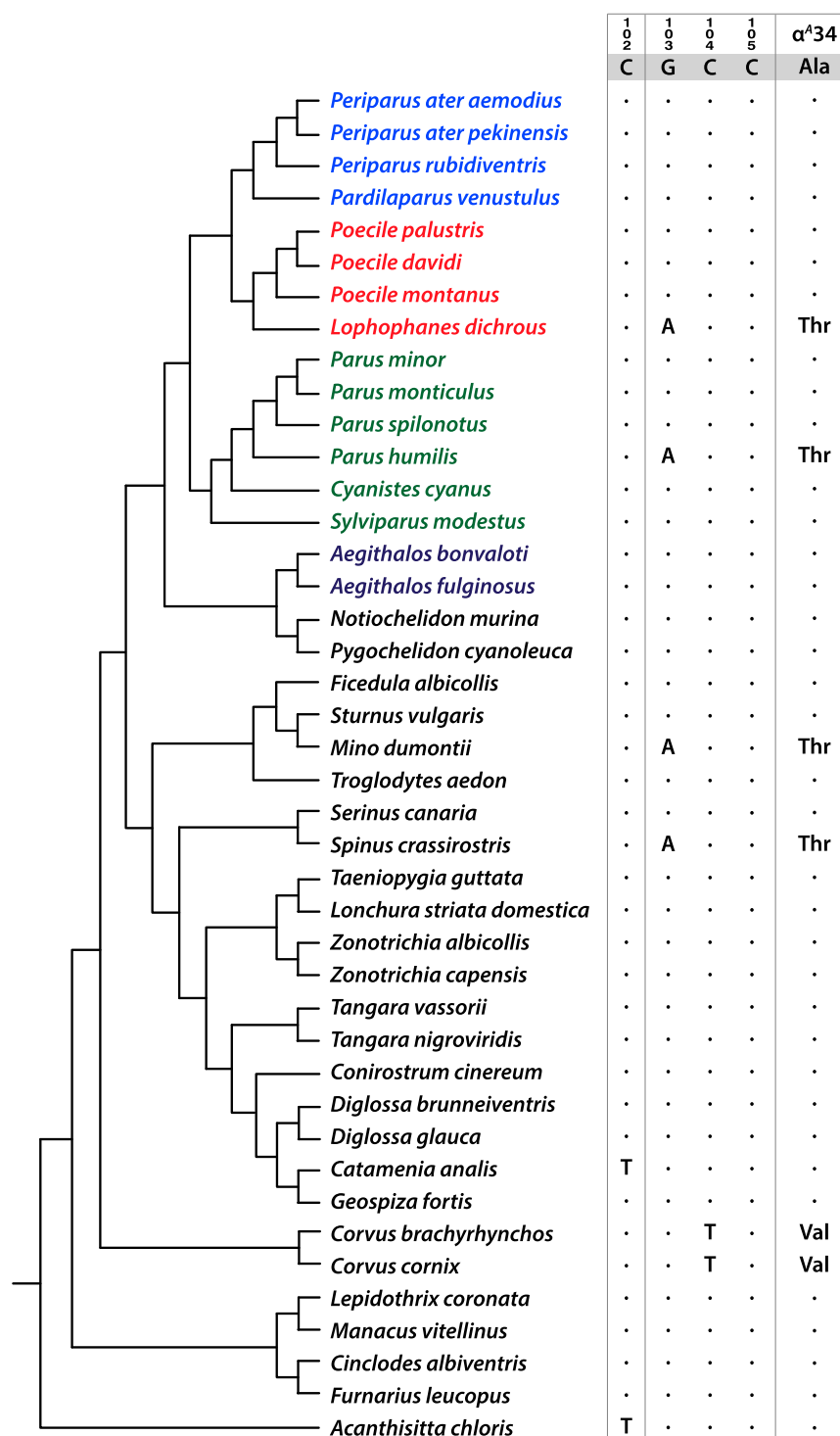


Fig. S4. Phylogeny of parid and aegithalid species (with color-coded species names), along with a diverse set of representative species from other passerine families. The nucleotide states to the right of the branch tips show recurrent nonsynonymous substitutions at the first and second codon positions of codon 34 in the α^A -globin gene. Parallel changes at the first codon position are attributable to substitutions at a CpG dinucleotide, CpG→CpA, which results in the replacement of Ala with Thr at α^{34} . The third position of the codon immediately preceding α^{34} is cytosine, C, in all but two species (*Catamenia analis* and *Acanthisitta chloris*), clearly indicating that the CpG dinucleotide represents the ancestral state for all oscine and suboscine passerines (i.e., the entire clade sister to *Acanthisitta chloris*). Thus, in the set of species shown here, the CpG dinucleotide has been eliminated multiple times independently, either by synonymous C→T substitutions at nucleotide site 102 (third position of the codon specifying residue 33) or nonsynonymous G→A substitutions at nucleotide site 103 (first position of the codon specifying residue 34). The high rate of parallelism at these two nucleotide sites is consistent with the high mutability of CpG sites relative to non-CpG sites.

Table S3. Predicted hydrogen bonds in the intradimer $\alpha_1\beta_1/\alpha_1\beta_1$ interface in deoxygenated (T) and oxygenated (R) states of avian Hb

Conformation	α^A34 amino acid state	Hydrogen bonds	Distance, Å
Deoxy (T)	Ala	Arg 31 [NH1] : Gln 127 [OE1]	2.66
		Arg 31 [NH1] : Phe 122 [O]	3.38
		Arg 31 [NH2] : Phe 122 [O]	3.36
		Gln 103 [OE1] : Gln 131 [NE2]	2.86
		Thr 111 [OG1]: Lys 120 [N]	3.52
		Leu 117 [O] : Arg 30 [NH2]	2.70
		Asp 126 [OD2] : Tyr 35 [OH]	2.86
	Thr	Gln-103 [NE2] : Asp-108 [OD1]	2.98
		Arg-31 [NH2] : Phe-122 [O]	2.66
		Thr-34 [OG1] : Pro-124 [O]	3.59
		Arg-31 [NE] : Gln-127 [OE1]	2.75
		Leu-117 [O] : Arg-30 [NH2]	2.82
		Asp-126 [OD2] : Tyr-35 [OH]	3.36
		Thr-111 [OG1] : Lys-120 [N]	3.52
Oxy (R)	Ala	Arg-31 [NH1] : Phe-122 [O]	3.05
		Leu-117 [O] : Arg-30 [NH2]	3.71
		Asp-126 [OD2] : Tyr-35 [OH]	3.16
		Tyr 36 [OH] : Gln 131 [NE2]	3.60
	Thr	Arg-31 [NE] : Gln-127 [OE1]	2.66
		Arg-31 [NH2] : Phe-122 [O]	2.76
		Gln-103 [NE2] : Asp-108 [O]	3.52
		Thr-111 [OG1] : Gly-119 [O]	2.66
		Gln-103 [OE1] : Gln-127 [NE2]	3.77
		Thr-111 [OG1] : Lys-120 [N]	3.69
		Leu-117 [O] : Arg-30 [NH1]	3.54
		Leu-117 [O] : Arg-30 [NH2]	3.25

Results are shown for two alternative Hb variants, those with the ancestral Ala α^A34 and those with the derived Thr α^A34 , as in *Parus humilis* and *Lophophanes dichrous*.